Hirotaka Ito

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Education

PhD	**Universitat Pompeu Fabra**, Economics Advisor: Professor Geert Mesters	2022–2027 (expected)
MRes	**Universitat Pompeu Fabra**, Economics	2021–2022
MS	**Barcelona School of Economics**, Economics	2020–2021

Research Visits

Brown University, Visiting Research Fellow — 2024
Host: Professor Toru Kitagawa

Research Interests

Primary field: Econometric theory

Secondary field: Applied econometrics

Topics: Partial identification, Bayesian inference, Nonparametric inference, Statistical decision theory

Job Market Paper

Asymptotic Minimax Optimality of Set Estimation

Abstract: Parameters of interest are often expressed as transformations of partially identified raw parameters. In such settings, the identified set of the targeted objects can be non-convex and there are no theoretical results on how well one can learn it from data. This paper fills part of this gap by studying the asymptotic efficiency of set estimation for a general class of identified sets. I first derive the asymptotic minimax efficiency bound for set estimators under risk functions based on the Hausdorff metric. A key ingredient in the analysis is a limit experiment framework tailored to the local structure of the identified set. I then show that a set estimator based on a criterion function is efficient under mild conditions. The analysis requires deriving the limiting distribution of the set estimator in the Hausdorff metric, which I obtain without globally characterizing the identified set via its support functions. Based on the limiting distribution, I propose a bootstrap-based confidence set. I illustrate my framework with an application to optimal taxation with behavioral agents.

Working Papers

Uniform and Locally Adaptive Quasi-Bayesian Inference for Conditional Moment Models

Abstract: This paper proposes a quasi-Bayesian method to construct confidence bands for a nonparametric function in conditional moment models. Importantly, we allow the true function to possess inhomogeneous smoothness over its domain, reflecting the locally varying complexity of the function of interest. Our confidence bands are constructed from quasi-Bayes posteriors with sieve GMM criterion functions and spike-and-slab priors. We show that the quasi-Bayes posteriors contract to the true function in the supremum norm at a rate that adapts to the unknown local smoothness. Based on this result, we derive a nonparametric Bernstein–von Mises-type result for the quasi-Bayes posteriors, in which the well-known covariance mismatch is avoided through the use of GMM criterion functions.

The resulting confidence bands are shown to achieve exact asymptotic coverage, while their widths, measured in the supremum norm, shrink at a nearly locally adaptive rate.

Teaching Experience

Barcelona School of Economics			
Advanced Econometric Methods I	Graduate (PhD)	TA	2022, 2023
Master's Thesis	Graduate (Master)	TA	2022
Universitat Pompeu Fabra			
Econometrics 1	Undergraduate	TA	2024, 2026
Econometrics 2	Undergraduate	TA	2022, 2023
Econometrics 3	Undergraduate	TA	2026
Introduction to Economics	Undergraduate	TA	2024

Conferences and Seminars

2024	BSE Summer Forum: Macroeconometrics and Policy Evaluation (Poster) Brown (Econometrics coffee, internal)
2023	BSE Summer Forum: Macroeconometrics and Policy Evaluation (Poster) UPF (internal)
2022	BSE Summer Forum: Advances in Econometrics (Poster) UPF (internal)

Awards and Scholarships

Barcelona School of Economics: Tuition Waiver 2020-2021

Skills

Languages: English (fluent), Japanese (native)

Programming: R, MATLAB, Python, Stata

References

Geert Mesters (advisor)
Economic Research Advisor
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Adam Lee
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